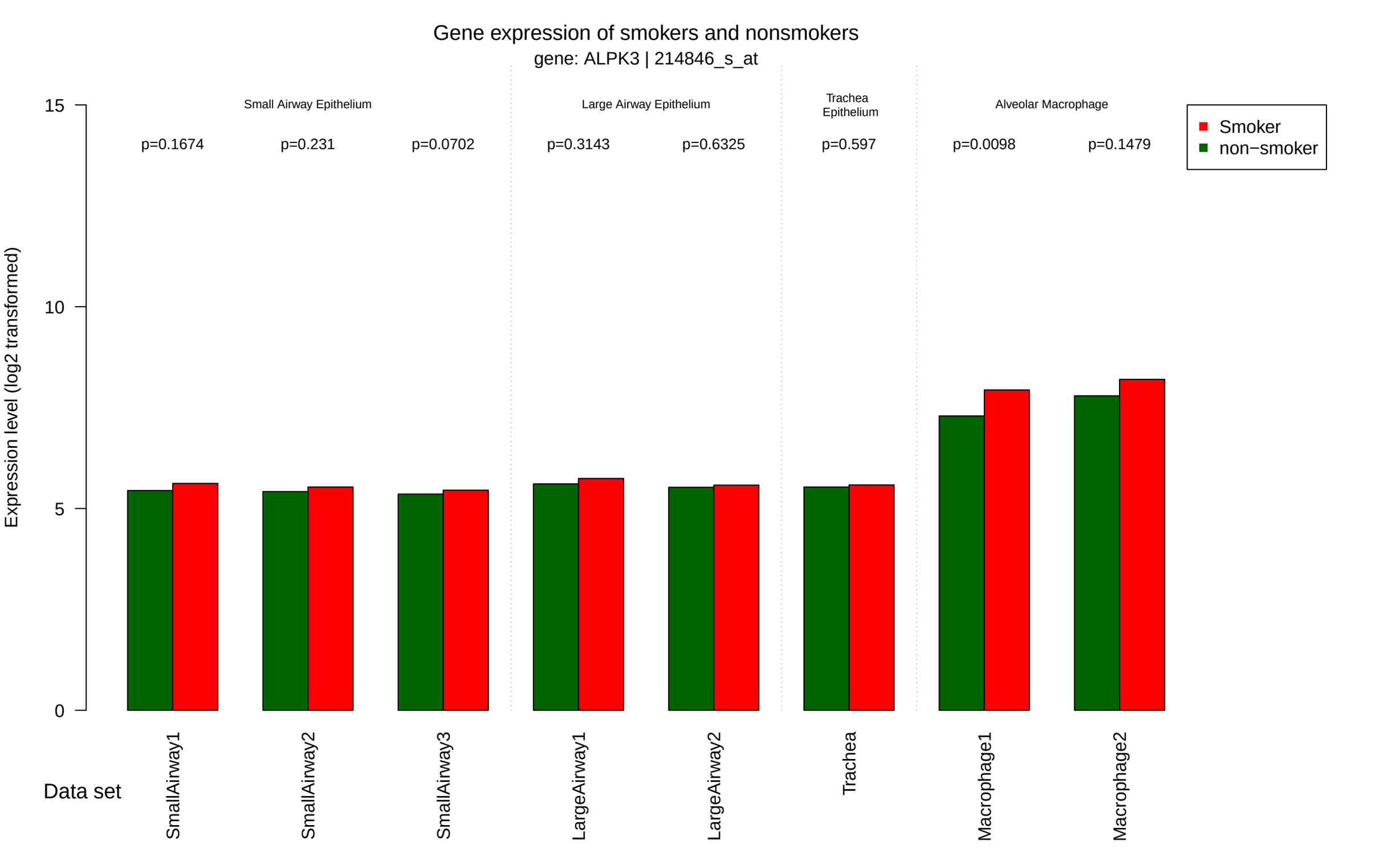
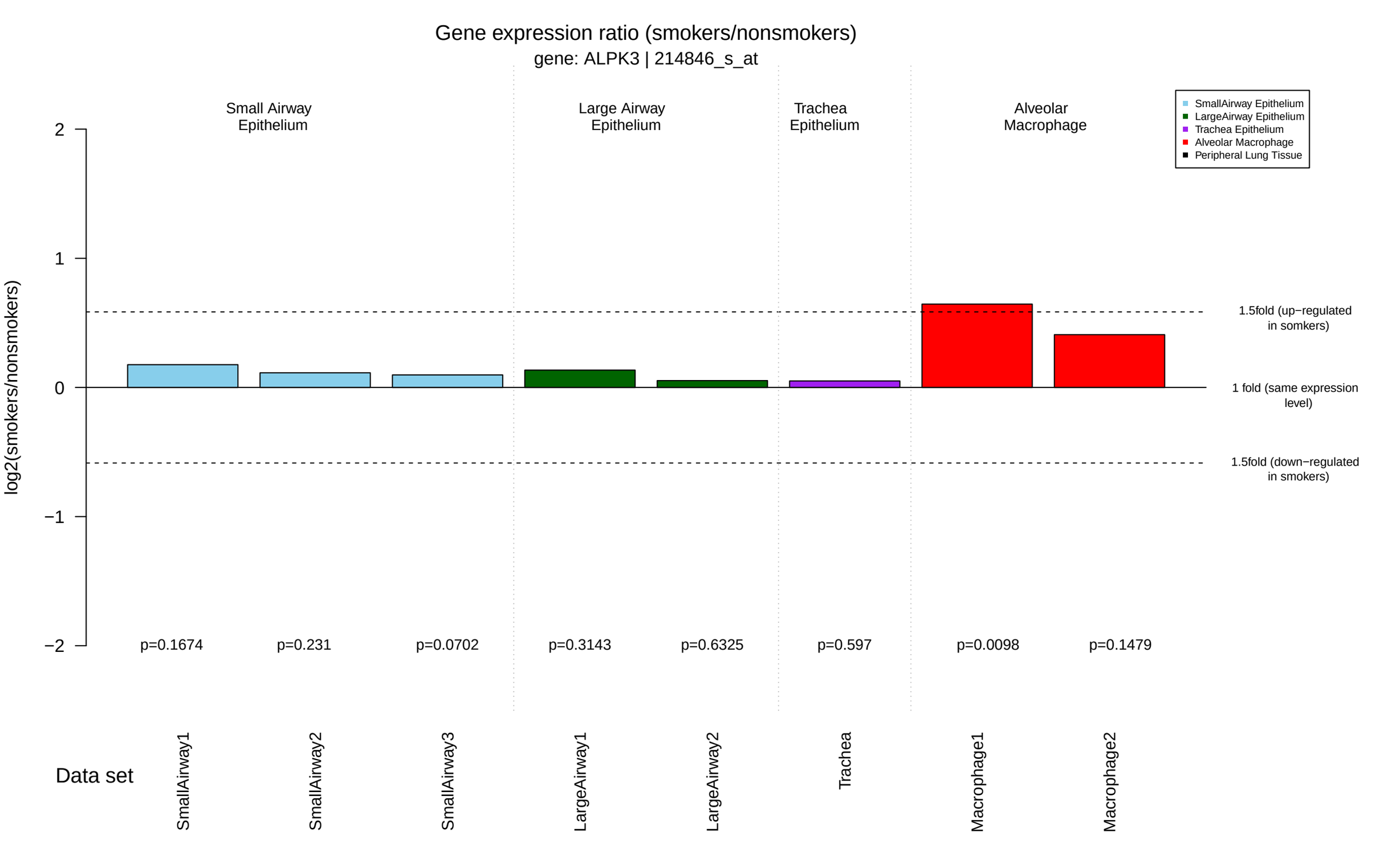
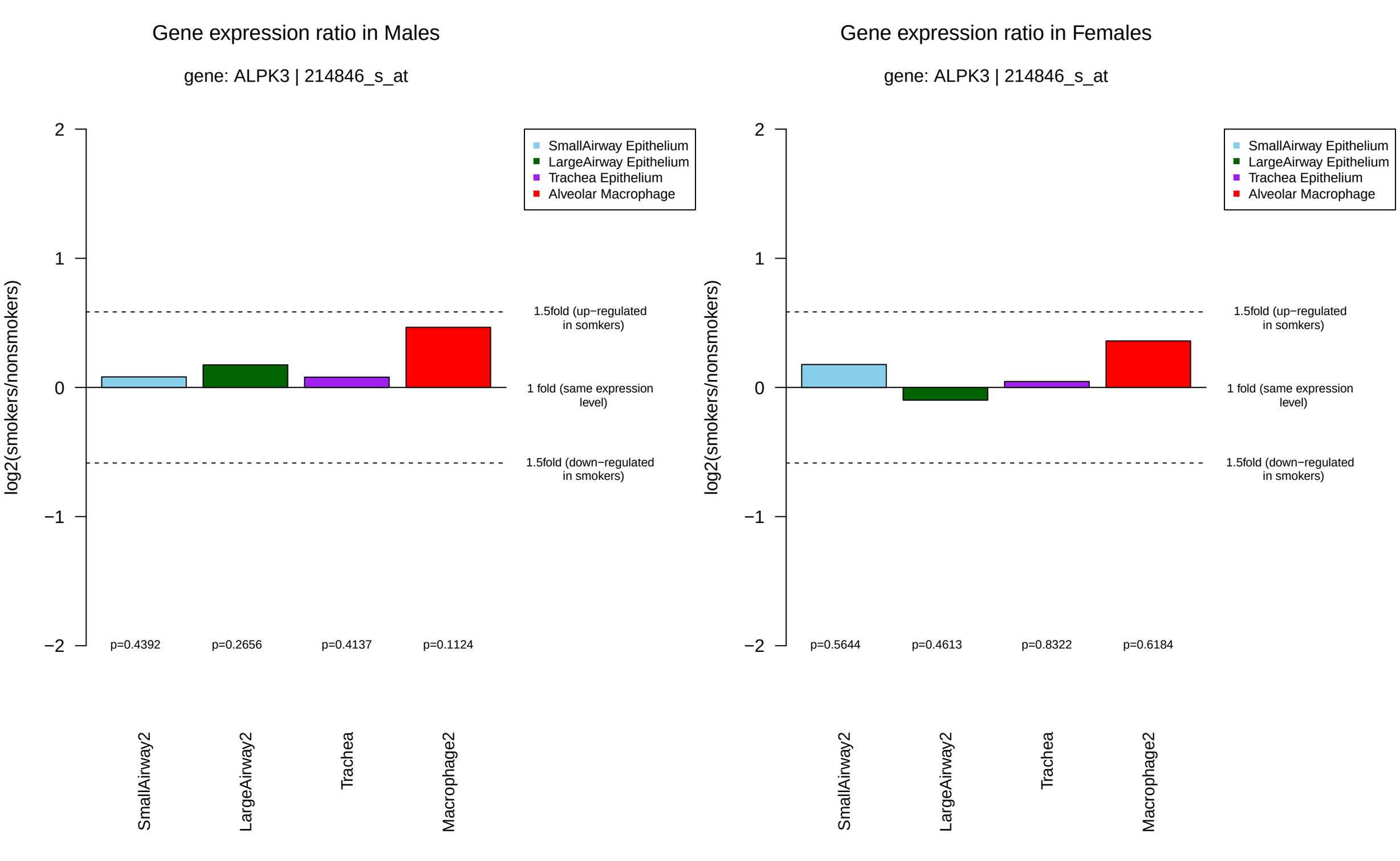
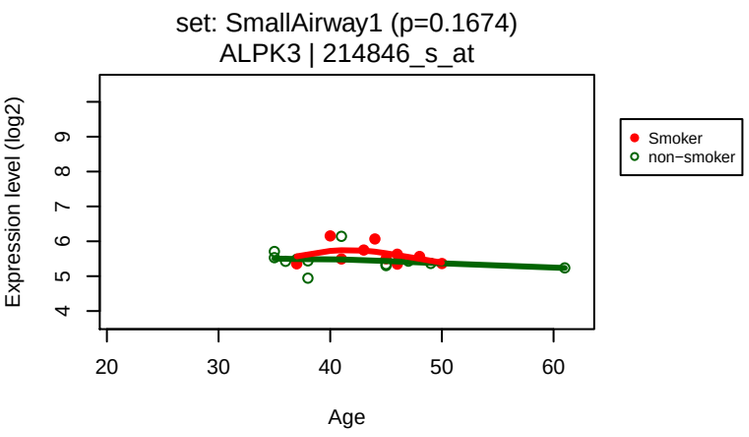
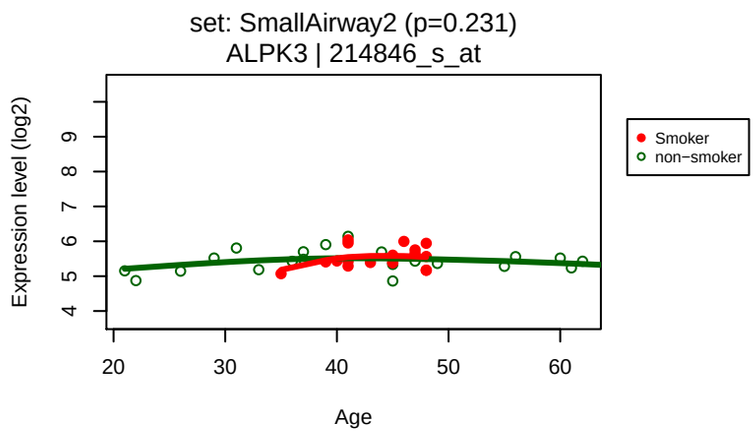
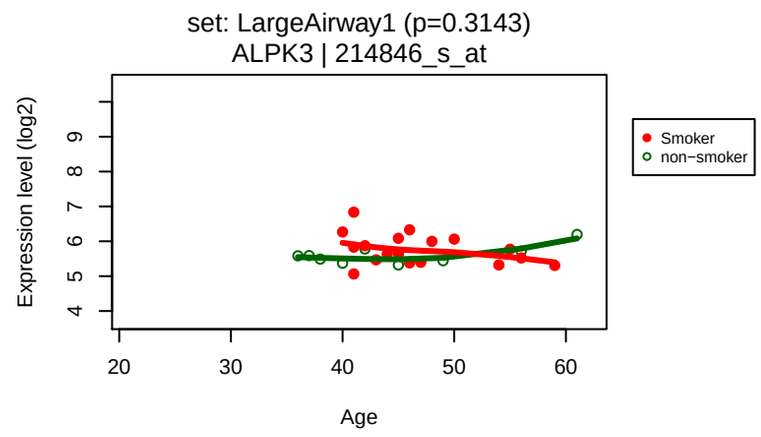

Gene expression of smokers and nonsmokers
gene: ALPK3 | 214846_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1674
p=0.231
p=0.0702
p=0.3143
p=0.6325
p=0.597
p=0.0098
p=0.1479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

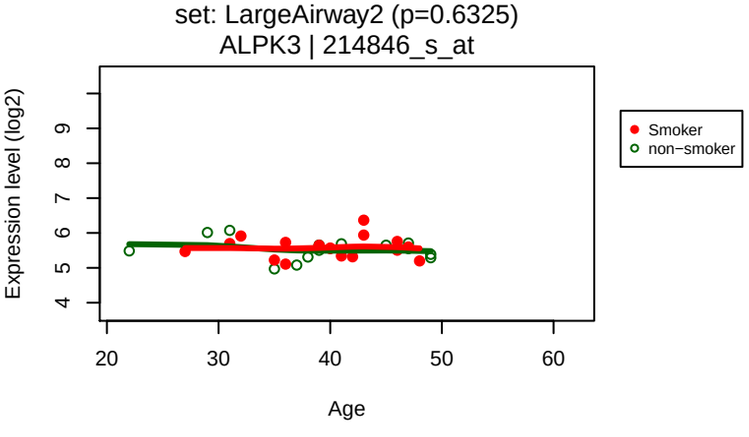

Gene expression ratio (smokers/nonsmokers)
gene: ALPK3 | 214846_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1674
p=0.231
p=0.0702
p=0.3143
p=0.6325
p=0.597
p=0.0098
p=0.1479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

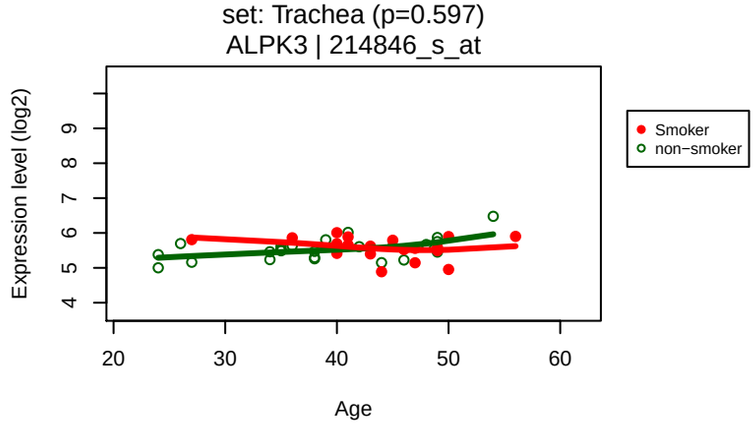

Gene expression ratio in Males
gene: ALPK3 | 214846_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4392
p=0.2656
p=0.4137
p=0.1124
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALPK3 | 214846_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5644
p=0.4613
p=0.8322
p=0.6184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

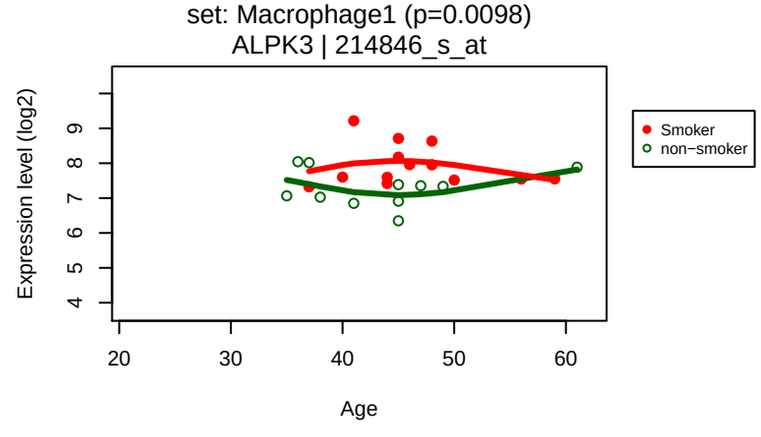

set: SmallAirway1 (p=0.1674)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker

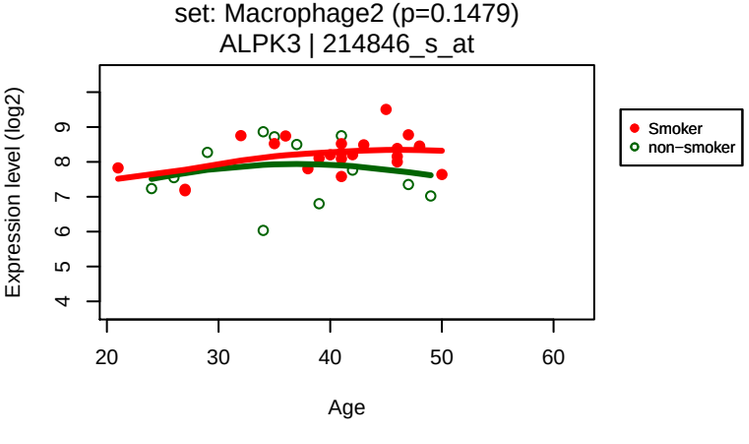

set: SmallAirway2 (p=0.231)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker

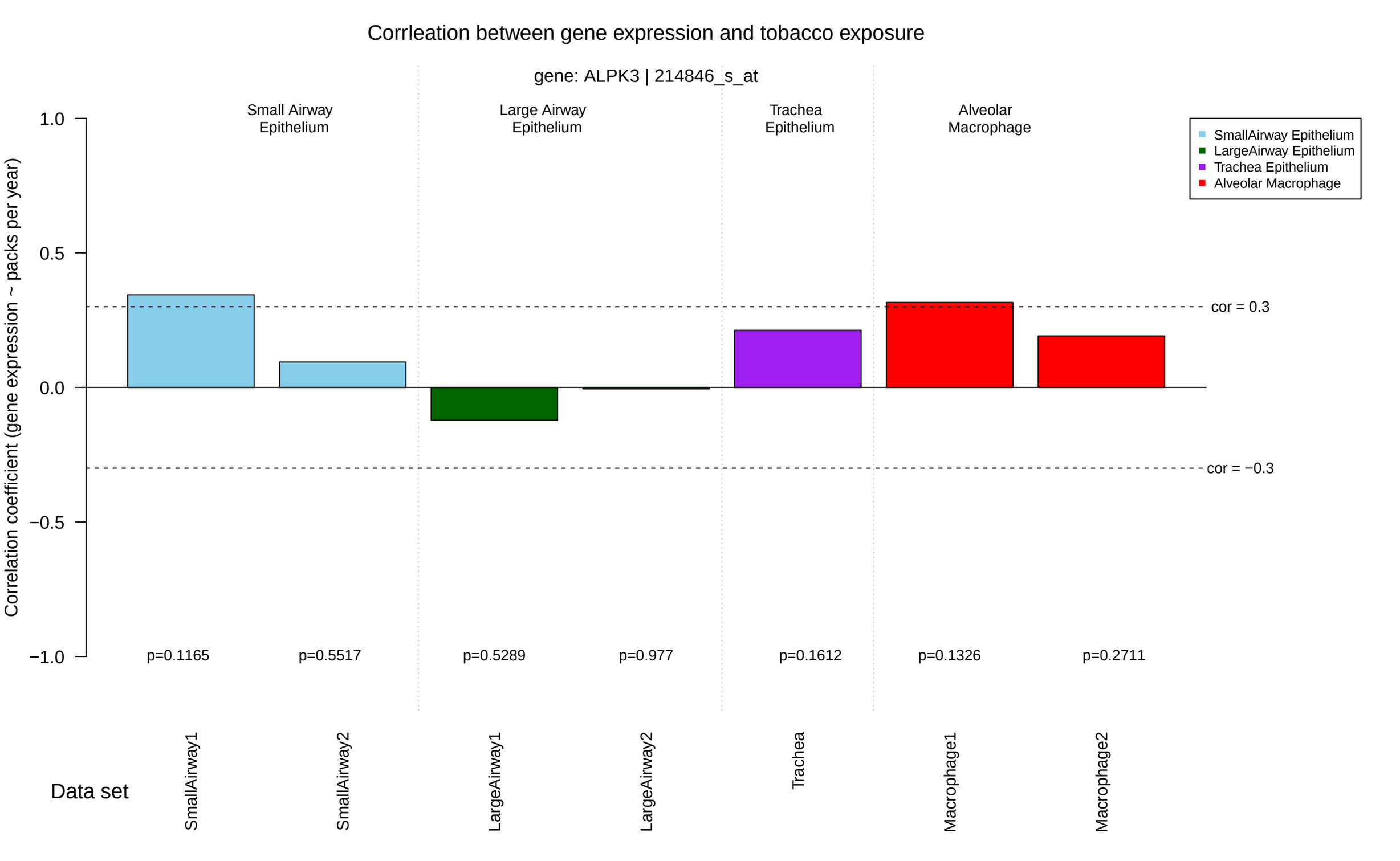

set: LargeAirway1 (p=0.3143)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6325)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.597)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0098)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1479)
ALPK3 | 214846_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALPK3 | 214846_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1165
p=0.5517
p=0.5289
p=0.977
p=0.1612
p=0.1326
p=0.2711
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2